Encoding	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R Type	Op-Code				Reg D			Reg S			Reg T			Function		
I Type	Op-Code				Reg D			Reg S			Signed Immediate					
J Type	Op-Code				Address											
LI Type	Op-Code				Reg D			S	Unsigned Byte							

Notes: 8 16-bit registers indexed by a 3-bit address, with separate instruction and data memory. R Type stands for register function work, I Type stands for signed immediate work, J Type stands for jump work, and LI Type stands for load immedate work (bigger numbers if I doesn't work). Op-Code provides 16 distinct commands, and decides instruction type.

Function	Ops	Effect	Use Cases
000	Add	RD = RS + RT	Adds Registers
001	Sub	RD = RS - RT	Subs Registers
010	And	RD = RS & RT	Masks Bits
011	Or	RD = RS | RT	Merges Bits
100	Xor	RD = RS ^ RT	Flips Bits / Compares Registers
101	Not	RD = ~RS	Inverts Register
110	Shift L	RD = RS << RT	Doubles RS RT Number of Times
111	Shift R	RD = RS >> RT	Halves RS RT Number of Times

Op-Code	Ops	Format	Effect	Use Cases
0000	**R** RD, RS, RT, Function	R	Function Effect	See Function Table
0001	**LI** RD, Hi/Lo, Byte	LI	Load Byte into Half of RD	Adds Byte into High/Low Byte of RD, based on Hi/Lo S bit
0010	**AddI** RD, RS, Imm	I	RD = RS + Imm (Signed, ±32)	Adds Signed 6-bit Imm into RS for counters (i = i + 1)
0011	**LW** RD, Imm(RS)	I	RD = dataMem[RS + Imm]	Locates dataMem Reg at base RS + offset Imm and loads register into RD
0100	**SW** RD, Imm(RS)	I	dataMem[RS + Imm] = RD	Locates dataMem Reg at base RS + offset Imm and loads RD into register
0101	**BEQ** RD, RS, Offset	I	If RD == RS, PC += Offset	Branches to offset PC value if RD = RS
0110	**BNE** RD, RS, Offset	I	If RD != RS, PC += Offset	Branches to offset PC value if RD != RS
0111	**BLT** RD, RS, Offset	I	If RD < RS, PC += Offset	Branches to offset PC value if RD < RS
1000	**J** Address	J	PC = Address	Jumps to Address PC value